Monte Carlo Simulation

Audity Ghosh

What is Monte Carlo Simulation?

- Monte Carlo Simulation is a way to solve problems using random numbers and probability.
- We simulate a process many times with random inputs and take the average of the results.
- It's useful when the problem is too complex to solve exactly.

Why is it called Monte Carlo?

- Named after the Monte Carlo Casino in Monaco, famous for games of chance.
- Just like roulette or dice, Monte Carlo simulations use randomness to model real-life situations.

Where is Monte Carlo Simulation Used?

- **Artificial Intelligence – *Training Algorithms***

Example:

Tuning hyper-parameters in machine learning (like learning rate).

🔹 Randomly try different settings.

🔹 Choose the one that gives best accuracy.

🔸 Faster than trying every possible combination.

How It Works – Step by Step

1. Define the Problem
2. Set Parameters and Ranges
3. Generate Random Values
4. Run the Simulation Many Times
5. Analyze the Results

Example: Estimating the Value of π

- A circle of radius 1 fits perfectly inside a square of size 2×2.
- The **area of the circle** is π.
- The **area of the square** is 4.
- If we randomly drop many points into the square and count how many fall **inside the circle**, we can use the ratio:

  π / 4 ≈ (Points inside circle) / (Total points)

Example: Estimating the Value of π

- 1. Randomly place points in a square ($0 \leq x, y \leq 1$)
- 2. Count how many fall inside a circle ($x^2 + y^2 \leq 1$)
- 3. $\pi \approx 4 \times$ (Points inside circle / Total points)
- As we use more and more points, this guess becomes more accurate!

Why It Works

- Monte Carlo uses the Law of Large Numbers:
- → More random trials = More accurate results
- It averages outcomes to estimate unknown values.
- **If you repeat a random experiment many times, the average result will get closer and closer to the expected value.**

Advantages and Limitations

- ✔ Advantages:
- - Simple and powerful
- - Works with complex systems
- - Easy to code

- ⚠️ Limitations:
- - Results are approximate
- - Requires many simulations
- - Relies on good random number generators

Viva Questions

- **What is Monte Carlo Simulation?**
  → A method to solve problems by running random simulations and averaging the results.
- **What happens when we increase the number of iterations?**
  → The result becomes more accurate because randomness evens out.
- **Why does Monte Carlo help estimate π?**
  → Because we can relate the ratio of points inside a circle to the total number of points to estimate π.
- **Where else is Monte Carlo used?**
  → In finance, weather forecasting, risk analysis, AI, and games.
- **Why are random numbers important in Monte Carlo?**
  → They simulate different possibilities. If the random numbers aren’t good, the results will also be bad.

Summary

- Monte Carlo Simulation helps solve complex problems using randomness.
- By simulating many random outcomes and averaging them, we can estimate answers.
- The more simulations we run, the closer we get to the real answer.